Population-Based RLVR: Breaking the Reweighting Ceiling via Independent Mode Discovery

Idea Sketch — Backburner

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The Problem: Single-Seed Mode Collapse

RLVR training is stochastic. Different random seeds find different solution modes.

But we only ever train **one policy**. That policy converges to **one mode** — whichever the random seed happens to find first. All diversity methods (DAPO, STEER, AEPO, DIVER, DSDR) try to preserve diversity within a single run. They slow mode collapse but cannot prevent it.

Single-seed RLVR:

- Seed 1 finds quicksort
- Converges hard to quicksort
- Mergesort pruned by step 100
- Pass@1 up, Pass@k down

What if we ran K seeds?

- Seed 1 finds quicksort
- Seed 2 finds mergesort
- Seed 3 finds radix sort
- Union coverage $>$ any individual

The question: can we **combine** the modes found by independent runs into a single, broader policy?

The Reweighting Ceiling (Recap)

From Yue et al. (2025): RLVR does not discover new reasoning patterns. It reweights within the base model's existing distribution.

$$\mathrm{Ceiling}(\pi_0) = \sup_{\pi \in \Delta(\pi_0)} \mathbb{E}_s[\mathrm{Pass@1}_\pi(s)]$$

where $\Delta(\pi_0) = \{\pi : \mathrm{supp}(\pi) \subseteq \mathrm{supp}(\pi_0)\}$.

All single-seed methods hit this ceiling. Diversity methods slow the approach but don't raise it.

Key insight from Yue et al.: distillation *can* introduce new knowledge and expand the reasoning boundary. A policy trained on another policy's solutions gains capabilities the base didn't have.

The Proposal: Population-Based RLVR

Phase 1: Divergent Training

$$\pi_{\theta_0} \xrightarrow{\text{seed 1}} \pi^*_{\theta_1} \quad \xrightarrow{\text{seed 2}} \pi^*_{\theta_2} \quad \xrightarrow{\text{seed 3}} \pi^*_{\theta_3} \quad \cdots \quad \xrightarrow{\text{seed K}} \pi^*_{\theta_K}$$

Phase 2: Solution Harvesting

$$\mathcal{D}_{\text{union}} = \bigcup_{k=1}^{K} \{(s,a) : a \sim \pi^*_{\theta_k},\ R_{\text{env}}(s,a) = 1\}$$

Phase 3: Cross-Pollination

$$\pi_{\theta_0} \xrightarrow{\text{SFT on } \mathcal{D}_{\text{union}}} \pi_{\theta_{\text{merged}}} \xrightarrow{\text{continue RLVR}} \pi_{\theta_{\text{final}}}$$

Each phase uses known techniques. The novelty is the **pipeline** and the claim that the union of independently discovered modes yields a strictly broader policy than any single run.

Why Independent Seeds Find Different Modes

GRPO's advantage estimation is stochastic: which rollouts happen to be correct in the first few batches determines the initial gradient direction.

Empirical evidence (anecdotal but consistent):

- Different seeds on the same problem set converge to different Pass@1 scores
- The *set* of problems solved differs across seeds — not just accuracy
- Seed A solves problems 1, 3, 5 that seed B cannot, and vice versa

Theoretical intuition: RLVR is a non-convex optimization with many local optima in policy space. Different initializations of the stochastic process (seed, batch order, sampling noise) converge to different optima. Each optimum corresponds to a different "solution mode" — a cluster of related strategies the policy has sharpened on.

Phase 1: Divergent Training

Train K copies of the base model independently via RLVR:

Design choice	Options	Default
Number of policies K	4, 8, 16	4 (budget-limited)
Differentiation source	Random seed only + different problem subsets + different hyperparameters	Seed only
Training budget per policy	Full budget / K Full budget each (expensive)	Full budget / K
Base model	Same for all	Qwen3-4B

Key tradeoff: Iso-compute (same total budget, split K ways) vs. $K\times$ compute. Iso-compute is the fair comparison — each policy trains for fewer steps but the total GPU-hours match a single long run.

Phase 2: Solution Harvesting

After divergent training, collect each policy's correct solutions:

For each problem s_i in the training set:

1. Sample N rollouts from each policy $\pi^*_{\theta_k}$
2. Keep all correct rollouts: $\mathcal{D}_k = \{(s_i, a) : R_{\text{env}}(s_i, a) = 1\}$
3. Union: $\mathcal{D}_{\text{union}} = \bigcup_k \mathcal{D}_k$
4. Deduplicate by solution strategy (embedding clustering)

Key measurement at this stage:

- Coverage: how many problems are solved by the union that no single policy solves?
- Diversity: how many distinct solution clusters exist in $\mathcal{D}_{\text{union}}$ vs. any $\mathcal{D}_k$?

Phase 3: Cross-Pollination

The critical step. How to merge K policies' knowledge into one:

Option A: SFT on $\mathcal{D}_{\text{union}}$, then continue RLVR

- SFT the *base model* (not any trained policy) on the diverse solution set
- This expands the base model's support to include all discovered modes
- Continue RLVR from this SFT checkpoint to sharpen

Option B: Forward-KL distillation from mixture

- Define mixture policy: $\pi_{\text{mix}}(\cdot|s) = \frac{1}{K}\sum_k \pi^*_{\theta_k}(\cdot|s)$
- Distill student via forward-KL: $\min_\theta D_{\text{KL}}(\pi_{\text{mix}} \| \pi_\theta)$
- Forward-KL is mass-covering — student must cover all modes (cf. DPH-RL)

Option C: Iterative cross-pollination

Option D: Teacher-Based Advantage from Population

GLM-5 introduced **on-policy cross-stage distillation**: instead of computing advantages from group-relative comparison, compute them from the teacher-student logit gap:

$$\hat{A}_{i,t} = \mathrm{sg}\left[\log \frac{\pi_{\theta_{\text{teacher}}}(y_{i,t} \mid x, y_{i,<t})}{\pi_{\theta}^{\text{train}}(y_{i,t} \mid x, y_{i,<t})}\right]$$

Tokens the teacher strongly prefers get high advantage. Tokens the student already matches get zero advantage. **No group comparison needed** — $G = 1$ suffices.

GLM-5 uses this temporally: each stage's checkpoint is a teacher for the next.

We can use it across the population: each of the K policies serves as a teacher for a fresh student. The advantage is the gap between what policy k would generate and what the student currently generates.

Why Teacher-Based Advantage Changes the Game

Standard GRPO cross-pollination:

- Harvest correct solutions from K policies
- SFT on union (off-policy, forgetting risk)
- Or distill via KL (mode collapse risk)
- Requires curating a dataset
- Two-phase: merge then train

Teacher-based cross-pollination:

- Each policy k acts as a teacher directly
- Advantage from logit gap (on-policy)
- Student generates, teacher evaluates
- No dataset curation needed
- Single phase: train with teacher signal

Key insight from GLM-5: $G = 1$ is sufficient when you have a teacher. The advantage comes from the teacher's logits, not from comparing rollouts within a group. This means the merged student can train with batch size 1024 $\times$ 1 rollout instead of 32 $\times$ 32 rollouts — massive throughput gain.

Multi-teacher extension: aggregate advantages from all K teachers. The student is pulled toward all K modes simultaneously. Weighting by per-problem teacher

Multi-Teacher Advantage Aggregation

Given K teacher policies, how to combine their signals:

Option D1: Mean teacher advantage

$$\hat{A}_{i,t} = \frac{1}{K} \sum_{k=1}^{K} \hat{A}_{i,t}^{(k)}$$

Simple but dilutes each teacher's specialization.

Option D2: Max teacher advantage

$$\hat{A}_{i,t} = \max_{k} \hat{A}_{i,t}^{(k)}$$

Student follows whichever teacher is most confident. Preserves specialization.

Why Forward-KL Matters for Merging

The choice of distillation divergence determines whether modes survive:

Reverse KL (standard RLVR):

- Mode-seeking
- Student covers one peak well
- Other modes dropped
- $\rightarrow$ Merging collapses back to one mode

Forward KL (DPH-RL style):

- Mass-covering
- Student must cover all teacher mass
- All modes preserved
- $\rightarrow$ Merging retains diversity

DPH-RL (2025) showed that forward-KL divergence preserves diversity in single-policy RLVR. We apply the same principle to multi-policy merging: distill from the mixture π_{mix} using forward-KL to ensure the student inherits all K modes, not just the dominant one.

Connection to EGSR

EGSR provides natural synergies with population-based training:

1. Dense evaluation of cross-pollinated solutions

When policy A's solutions are SFT'd into policy B, EGSR's teacher can evaluate them with failure-mode granularity — not just `did it pass''` `buthow` does this new strategy compare to the existing ones."

2. Efficient per-policy training

Each of the K policies trains with EGSR's dense reward, reaching its local optimum faster. If EGSR reduces the group size needed (EGSR G=4 $\approx$ GRPO G=16), the per-policy budget goes further.

3. Judgment diversity across policies

GLM-5 vs. Our Proposal: Temporal vs. Parallel Population

	GLM-5	**Ours**
Population type	Temporal (sequential stages)	Parallel (K seeds)
Diversity source	Different objectives per stage	Stochastic optimization
Teachers	Prior checkpoints	Concurrent peer policies
Advantage	Single teacher logit gap	Multi-teacher aggregation
G per prompt	1 (teacher provides signal)	1 (same insight applies)
Compute	$1\times$ (sequential)	$K\times$ or iso-compute
Mode diversity	Limited (shared trajectory)	Higher (independent seeds)
Forgetting	Addressed by recovery stage	Avoided by parallel training

GLM-5 validates the **mechanism** (teacher-based advantage for cross-pollination). We propose a different **topology** (parallel population rather than sequential pipeline) that should yield more diverse modes because the policies never share training trajectory.

Prior Work Landscape

Paper	Approach	Gap from ours
PBT (Jaderberg 2017)	Population-based training	Classical RL, not LLMs
Yue et al. (2025)	RLVR = reweighting	Diagnosis only, no population
DPH-RL (2025)	Forward-KL for diversity	Single policy, no population
DIVER / DSDR / AEPO	Diversity rewards	Single policy, within-run
POPE (CMU 2026)	Oracle-guided exploration	External oracle, single policy
GLM-5 (2025)	Cross-stage distillation via teacher advantage	Temporal, not parallel; same trajectory, less mode diversity
BroRL (NVIDIA)	Scale rollout breadth	Single policy, no merging
Ensembles of distilled	Ensemble for generalization	Classical RL, not LLM RLVR

The gap: Parallel multi-seed RLVR with teacher-based cross-pollination. GLM-5 validates the teacher advantage mechanism but uses it temporally. We use it across a parallel population for greater mode diversity.

Experimental Plan

Phase 0: Mode divergence validation (1 week)

- Train K=4 RLVR runs (Qwen3-4B, different seeds) for 200 steps each
- Measure per-run coverage, pairwise overlap, union coverage
- **Go/no-go:** if union coverage meaningfully exceeds any individual ($>5% more problems solved), proceed

Phase 1: Cross-pollination comparison (4 weeks)

- Option A: SFT base on $\mathcal{D}_{\text{union}}$, continue RLVR
- Option B: Forward-KL distillation from π_{mix}
- Option D: Teacher-based advantage with competence weighting (G=1)
- Compare all against: single run (same total compute), single run ($K\times$ compute), single run with diversity bonus

Phase 2: Analysis (2 weeks)

The Iso-Compute Question

The fairest comparison is **iso-compute**: same total GPU-hours.

Method	Total compute	Prediction
Single RLVR, 800 steps	C	High Pass@1, low diversity
K=4 RLVR, 200 steps each	C	Lower per-policy Pass@1
+ cross-pollination	$C + \epsilon$	Higher coverage?
Single RLVR + diversity bonus	C	Moderate both

The key hypothesis: K short runs + merge $>$ 1 long run, even at the same compute budget.

Why this might hold: 200 steps is enough to identify a mode (early training). The last 600 steps of a long run are pure sharpening — diminishing returns. Spending those 600 steps finding 3 other modes, then merging, could yield more total coverage.

What Could Go Wrong

Risk	Severity	Mitigation
Seeds find the same mode	**High**	Measure in Phase 0; diversify problem subsets per seed
SFT merging forgets modes	Medium (reduced)	Teacher-based advantage bypasses SFT entirely (GLM-5)
Per-policy undertrained (iso)	Medium	Find minimum viable steps per policy (ablate)
$K\times$ compute not justified	Medium	Iso-compute comparison; G=1 throughput gain helps
GLM-5 temporal approach is sufficient	Medium	Phase 0 measures parallel vs temporal diversity directly

Kill criterion: Phase 0 shows $<$3% union coverage gain over best individual. Seeds aren't diverging enough to justify the pipeline. Secondary kill: parallel population doesn't yield more diversity than GLM-5-style sequential stages.

Theoretical Angle

Coverage amplification: Let C_k be the set of problems solved by policy k. If policies are trained independently:

$$\mathbb{E}\left[|\bigcup_k C_k|\right] \geq |\mathbb{E}[C_1]|$$

with equality only if all policies find identical solutions. Under diversity (different seeds), the union grows sublinearly but strictly faster than a single run.

Connection to the reweighting ceiling: Cross-pollination via SFT expands $\mathrm{supp}(\pi_\theta)$. If policy 2's solution was not in policy 1's support, SFT on it literally raises policy 1's reweighting ceiling. Subsequent RLVR then optimizes within this expanded support.

Formal claim:

Honest Assessment

Pros:

- Conceptually clean: independent exploration + unified exploitation
- No architectural changes, no external oracle
- Directly addresses the reweighting ceiling with a clear mechanism
- GLM-5 **validates the merging mechanism** (teacher-based advantage works)
- Teacher-based advantage enables $G{=}1$ — massive throughput gain for merging
- **Least crowded** of the five exploration ideas

Cons:

- $K\times$ compute (even iso-compute means shorter per-run training)
- Phase 0 could kill it: seeds might just find the same mode